

First Webinar: How to be proficient in the ESPSS

1st March 24

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- **Introduction**
 - **New type of Webinars, dedicated to toolkits**
 - **First ESPSS webinar**
 - **3-4 ESPSS webinars in 2024**
 - **Objective:**
 - **Show key features by using examples in a short time**
 - **All ESPSS webinars will be recorded and uploaded to a cloud for users with the maintenance up to date**
 - **1st Webinar**
 - **General Introduction to ESPSS toolkit**
 - **Future Webinar**
 - **Optimize Propulsion models, debugging...**
 - **Based on interest and feedback**



- **Hot to be Proficient in ESPSS**

- 1. Setting Up Simulations**

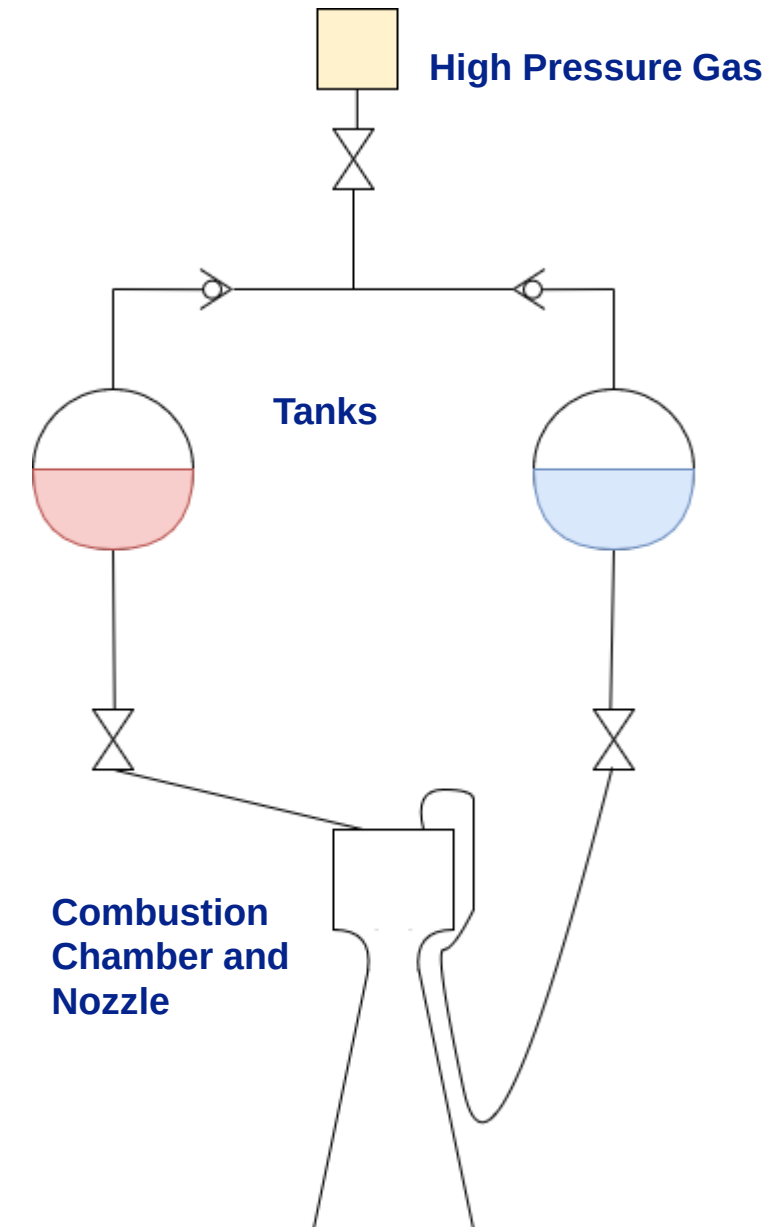
- **Choose the Working Fluids**
 - **Select the proper components of the simulation**
 - **Fluid Elements**
 - **Thermal Elements**
 - **Control Elements**
 - **Robust Initialization**
 - **Other Tips**

- 2. Debugging**

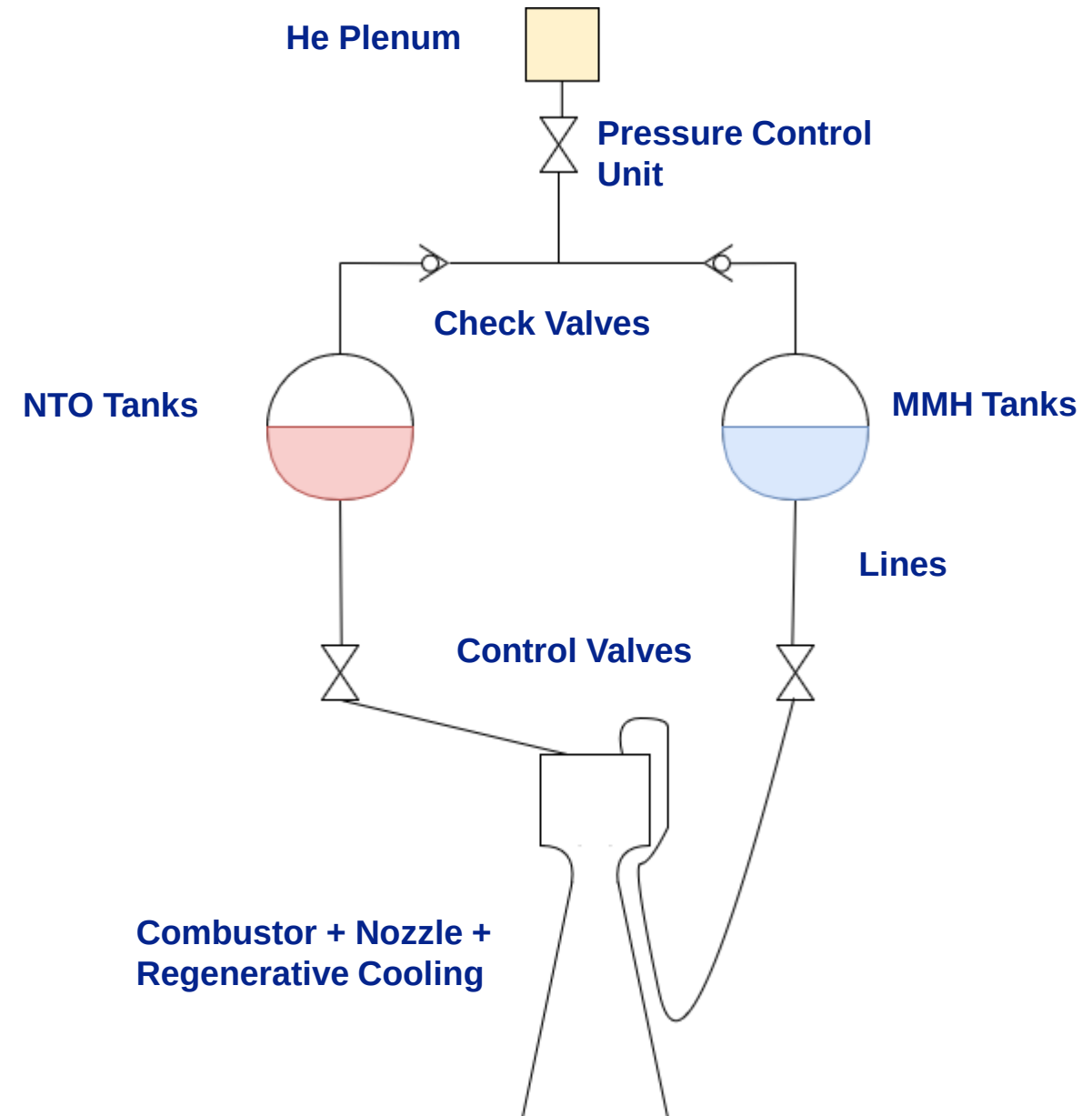
- **Boundaries**
 - **Using the Monitor**
 - **Post-Processing**

Setting Up Simulations

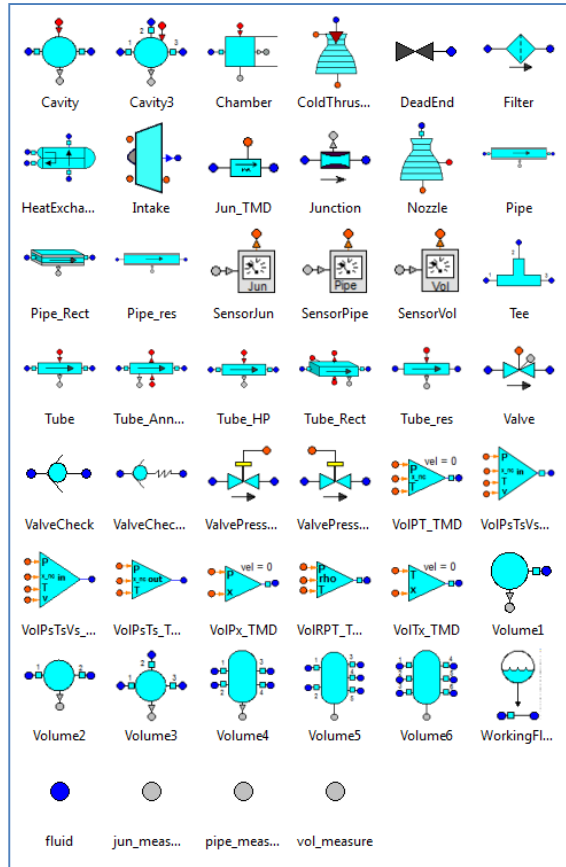
- **Specification**
 - **Pressure Fed MMH-NTO Engine**
 - **Design Given**
 - **Helium Pressurizer**
 - **Cooling Jacket**
 - **Physical Effects**
 - **Vaporization of fuels in tanks**
 - **Two phase flow evolution in cooling jacket**
 - **Transient Evolution in the chamber**



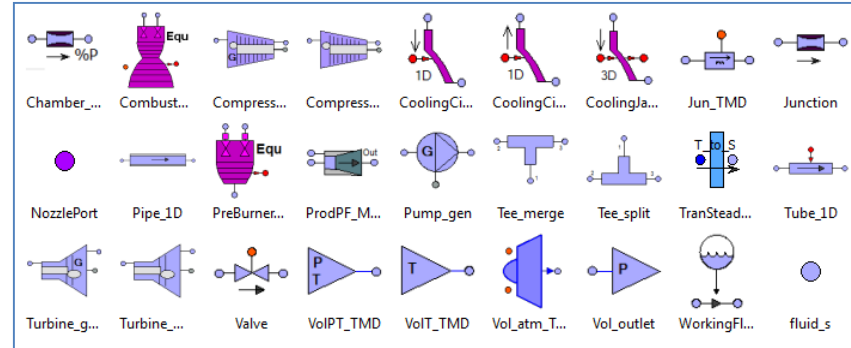
- **Working Fluids**
 - Liquid and Gas MMH, NTO
 - Gas Helium
- **Components**
 - Two Phase tanks for MMH and NTO
 - Single Phase Tank for He
 - Valves
 - Pressure Regulator Valve
 - Check Valves
 - Injector Valves
 - Lines
 - Cooling Jacket
 - Tees



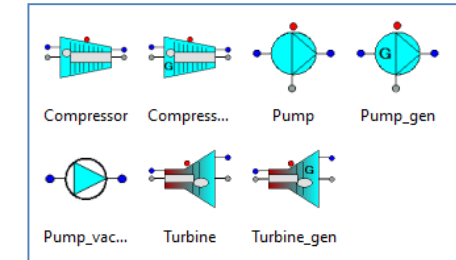
FLUID_FLOW_1D



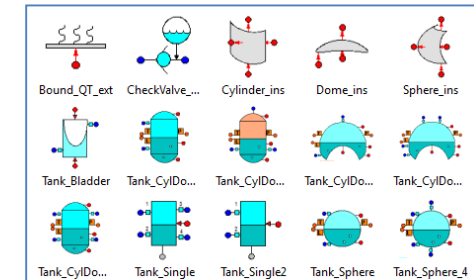
STEADY



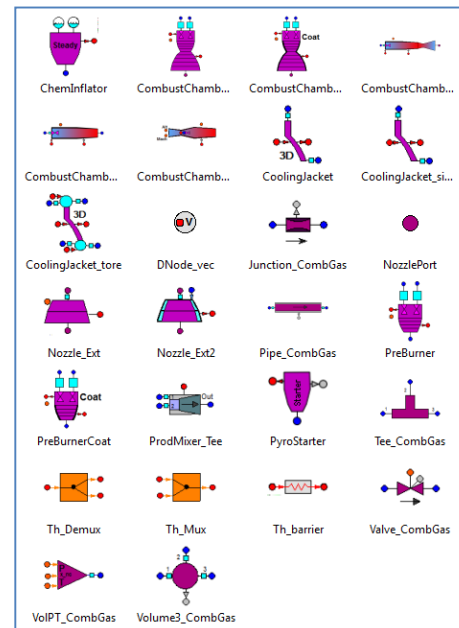
TURBOMACHINERY



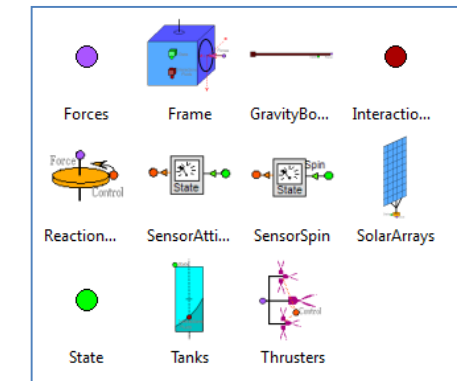
TANKS



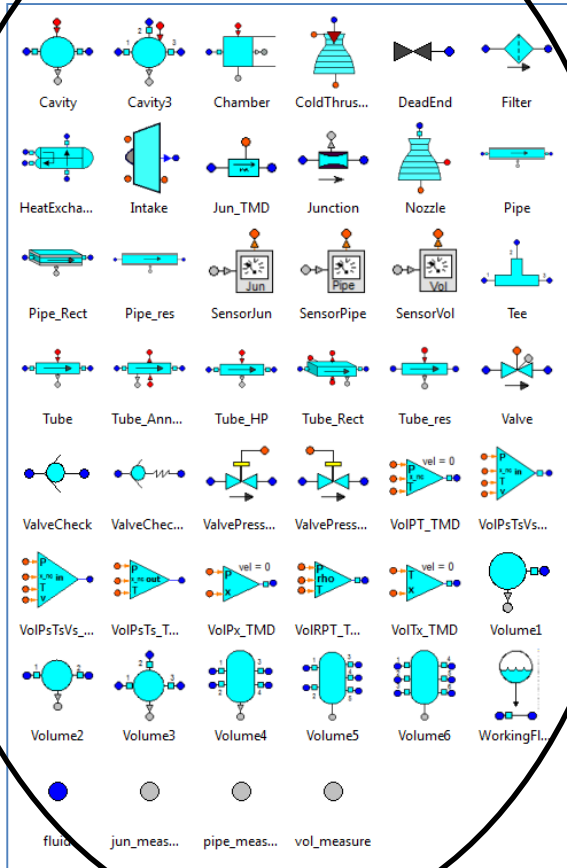
COMB_CHAMBERS



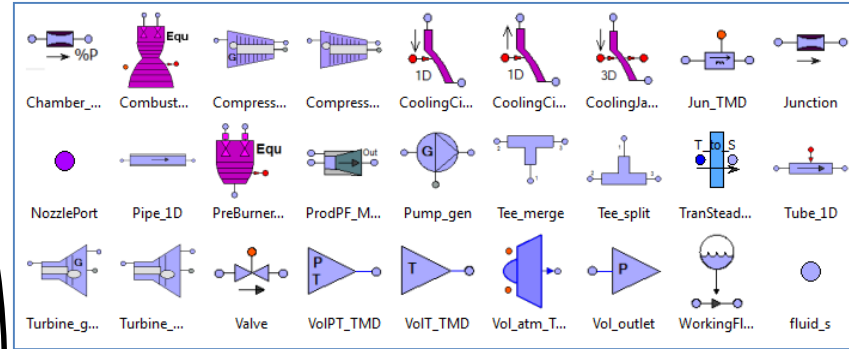
SATELLITE



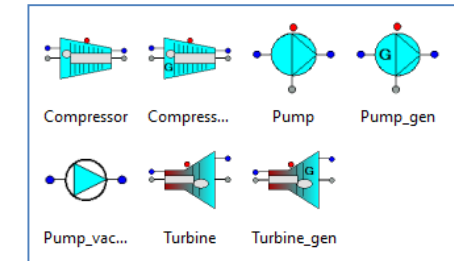
FLUID_FLOW_1D



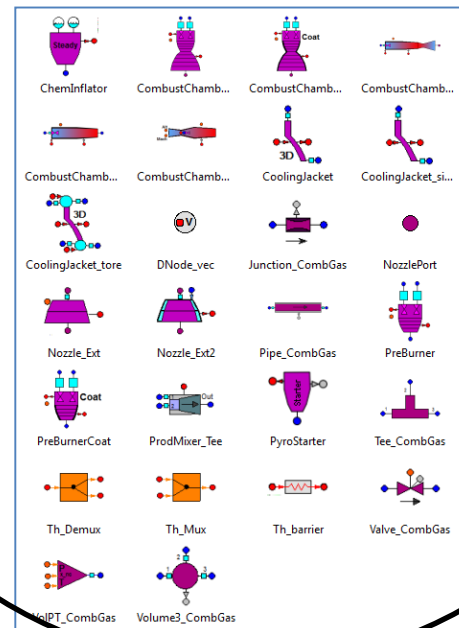
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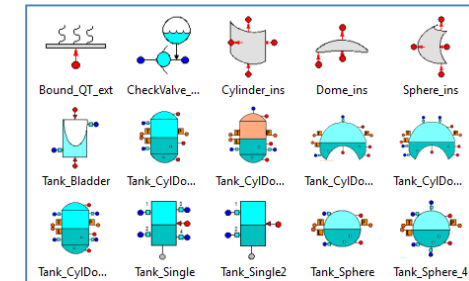
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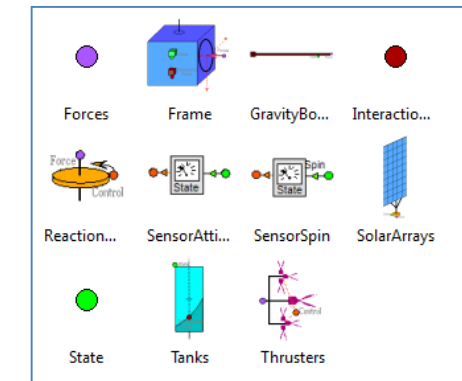
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TANKS



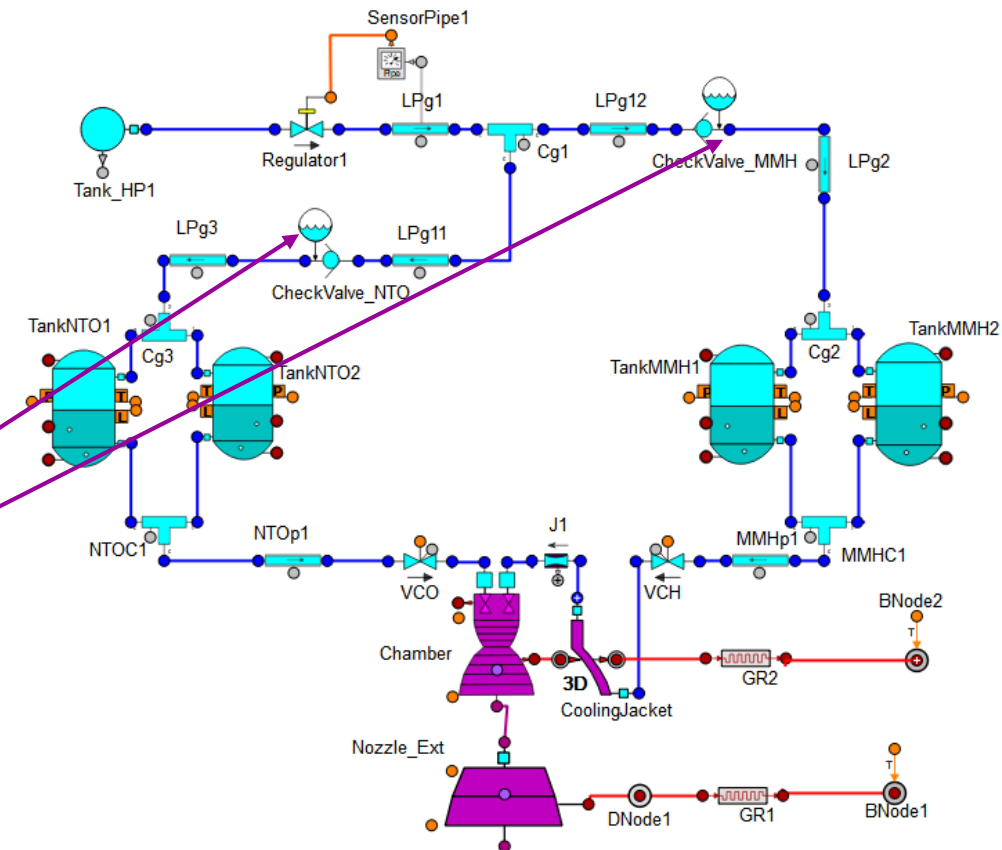
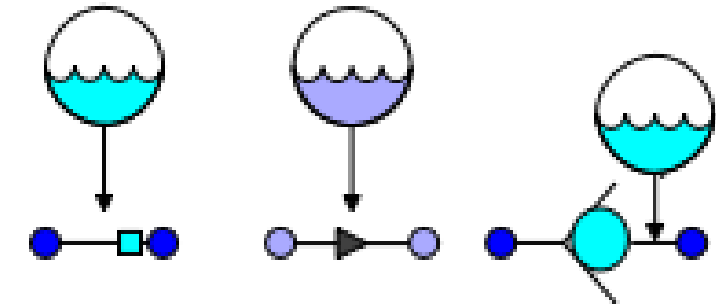
SATELLITE





- **Working Fluids**
 - **Pure fluids**
 - Real fluid properties interpolated in external 2D property files (mostly obtained from the REFPROP code (NIST))
 - Perfect gas properties interpolated in external properties files
 - Simplified liquids interpolated in external temperature-dependent properties files
 - Van der Waals functions
 - **Mixtures (main fluid and non-condensable fluid)**
 - Perfect gases
 - Real fluid – Perfect gas
 - Real fluid – Real gas

- Working Fluids
 - FF_1D
 - One main fluid and non-condensable fluid
 - STEADY
 - One main fluid
 - TANKS (CheckValve_FISep)
 - Allows having one pressurizer for several main fluids
 - Downstream of Burners (CC, STEADY)
 - Mixture of perfect gases
 - Our case study
 - Two phase flow evolution of fuels
 - He Feeding , same tank for both fuels
- ↓
- Non-Condensable He (Pf or Real)
 - Real MMH and Real NTO



- Tanks

- Cavity

- One node, “constant” volume HEM. Level is not considered in ports

- Chamber

- One node, variable volume.

- Cryo Tanks

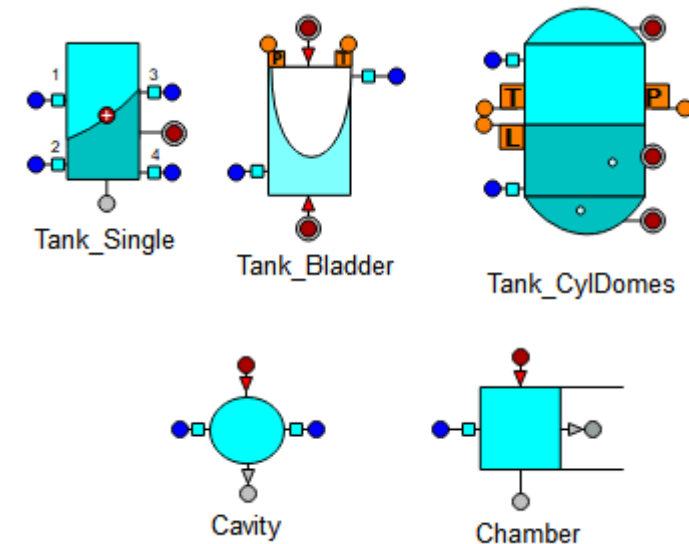
- Multiple nodes for liquid, gas (1D) and walls (2D)
 - Film Boiling, Vaporization, Condensation, Stratification

- Bladder Tanks

- Two chambers for liquid and gas with total constant volume

- Single Tanks

- Advanced Cavity with overflow and sloshing calculations



- Tanks

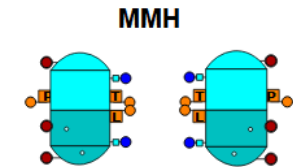
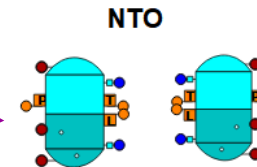
- Pressurizer

- Is Stored Independently (only one fluid)
 - Cavity is decided to be used



- Reducer and Oxidizer

- Stored with pressurizer (two fluids)
 - Liquid can vaporize inside the tank
 - Cryo Tank has to be used
 - 2 Tanks for the NTO and 2 for the MMH



- **Valves and Lines**

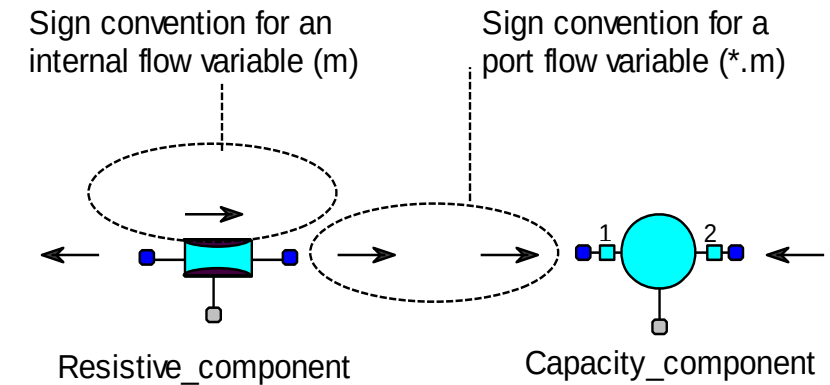
- **Resistive Components and Capacitive Components**

- Resistive components calculate mass flow
- Capacitive components calculate state ($\rho, u, P, T \dots$)

- **Valves and Junctions are resistive components**

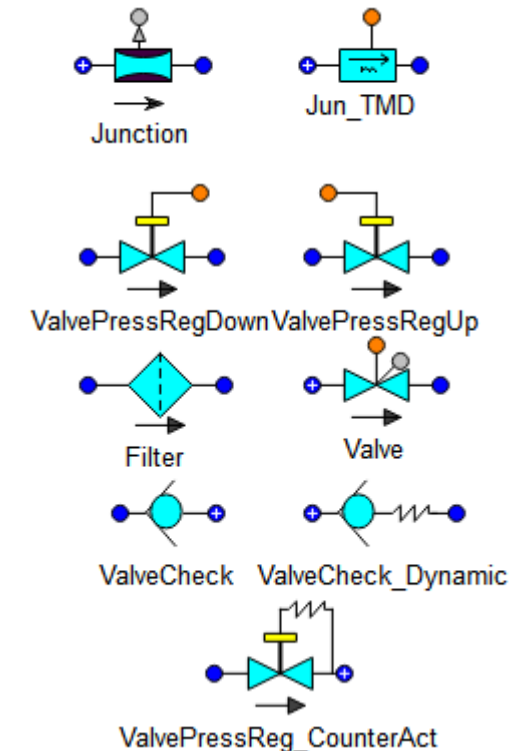
- **Lines, accumulators and Tanks are capacitive components**

- Resistive Lines are available, if needed



• Resistive Components

- Junction component to be used for connection between elements
 - $m(\text{Area} = \text{cte}, \zeta)$
- Filter
 - $m\left(\frac{\rho}{\rho_{\text{ref}}}, \frac{\mu}{\mu_{\text{ref}}}, \frac{dp}{dp_{\text{ref}}}\right)$
- Jun_TMD
 - Imposed mass flow. Remove pressure dependence.
- Valve
 - Area function of Position
- ValvePressRegDown
 - Valve with Position $f(P, P_{\text{down}})$
- ValvePressRegUp
 - Valve with Position $f(P, P_{\text{up}})$
- ValveCheck
 - Valve with Position $f(dp, dp_{\text{min}}, dp_{\text{max}})$
- ValvePressReg_CounterAct, ValveCheck_Dynamic
 - Valves with position $f(\text{mechanical actuator})$



- Capacitive Components

- Tube and Pipe (Length, multiple nodes)

- Friction, Bends

- Geometry

- Tube

- D

- Tube_React

- a,b

- Tube_Annular

- A_cr, P_in, P_out

- Tube_HP

- A_cr, P_out

- Tube

- Thermal port for the walls (T is boundary)

- Pipe

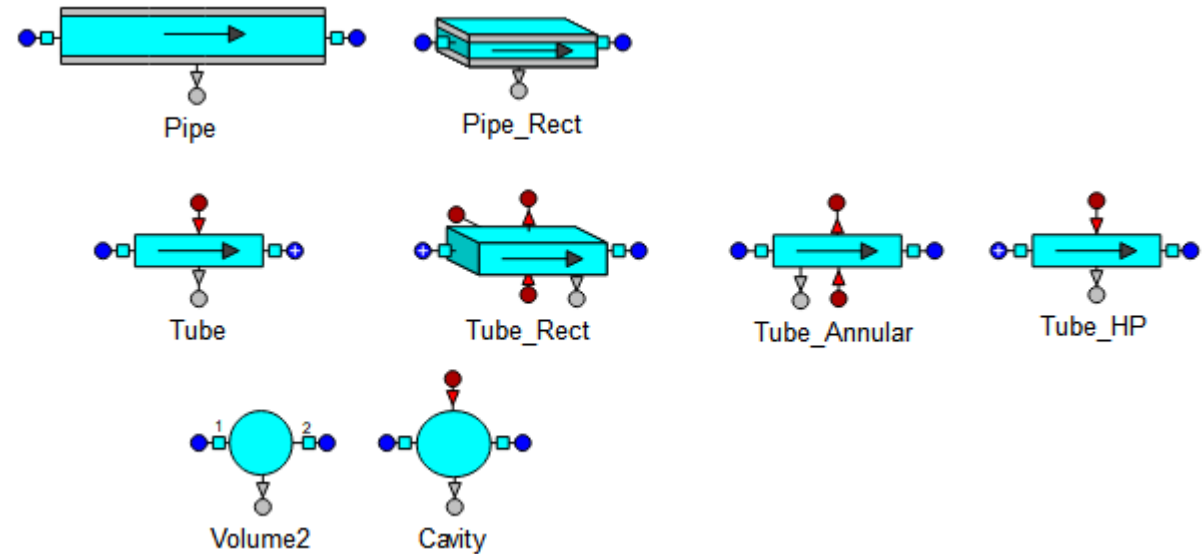
- Includes Wall and wall expansibility, with heat exchange to a constant ambient temperature

- Volume, Cavity (0D)

- No friction

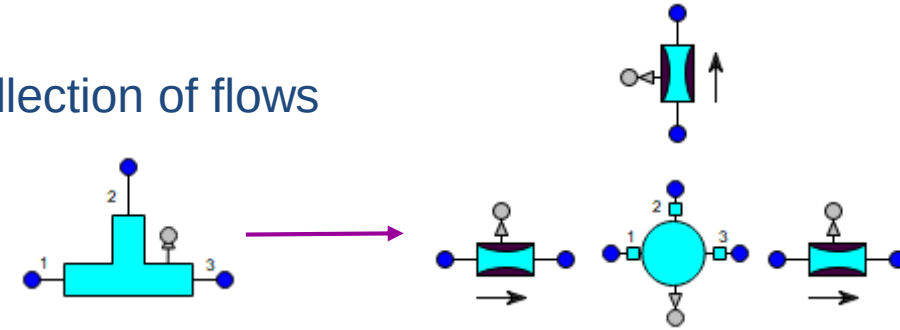
- Vol is adiabatic

- Cavity has thermal port (like TUBE)

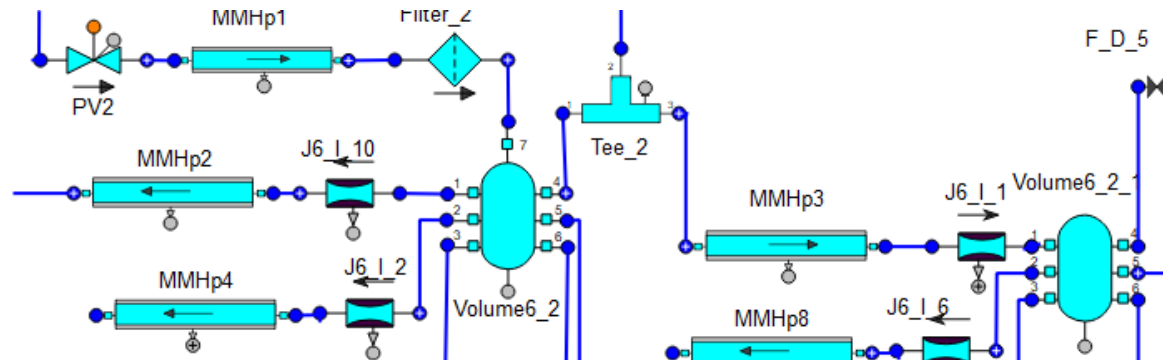


MANIFOLDS – JUNCTIONS

- Use the **Tee** component for single pipe bifurcation or collection of flows



- Volume** components are also representative of fluid capacities mixing or splitting flows depending on the conditions of connected components
- For connections involving more than three lines, use a **Volume** and put a junction (or a control valve) between the volume and the pipes



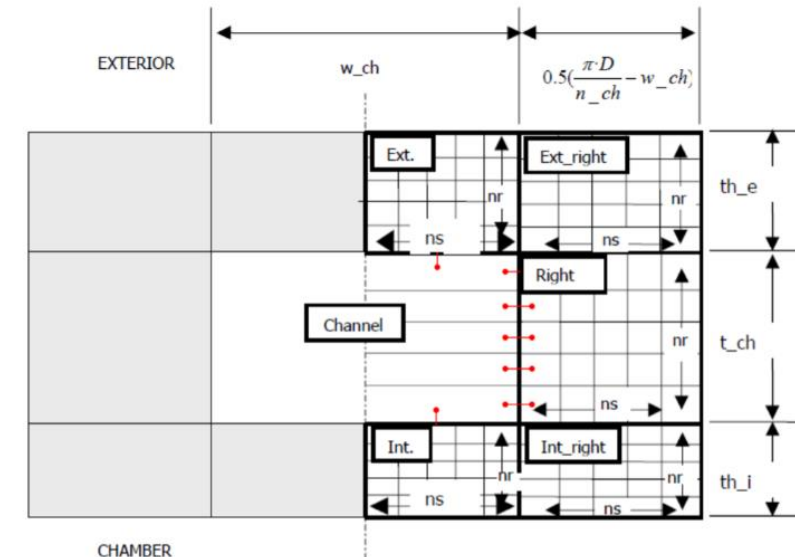
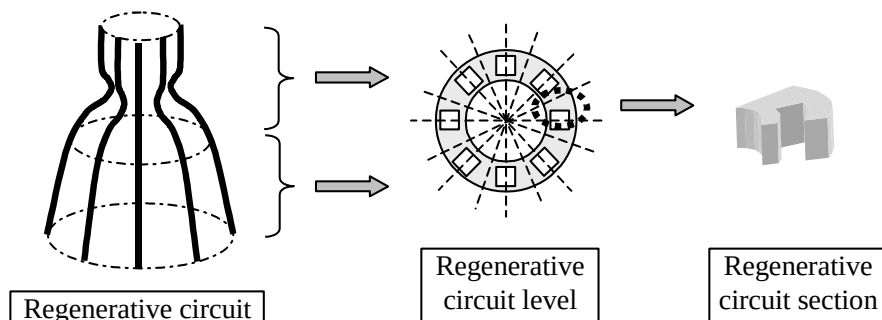
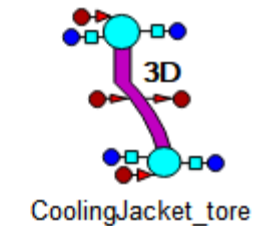
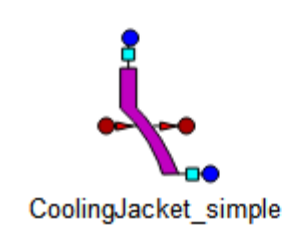
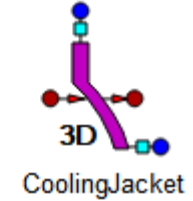
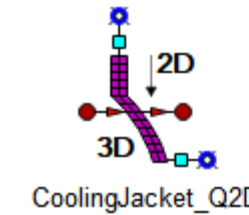
- Use the **DeadEnd** component to close any fluid port that will not be used



• Cooling Jackets

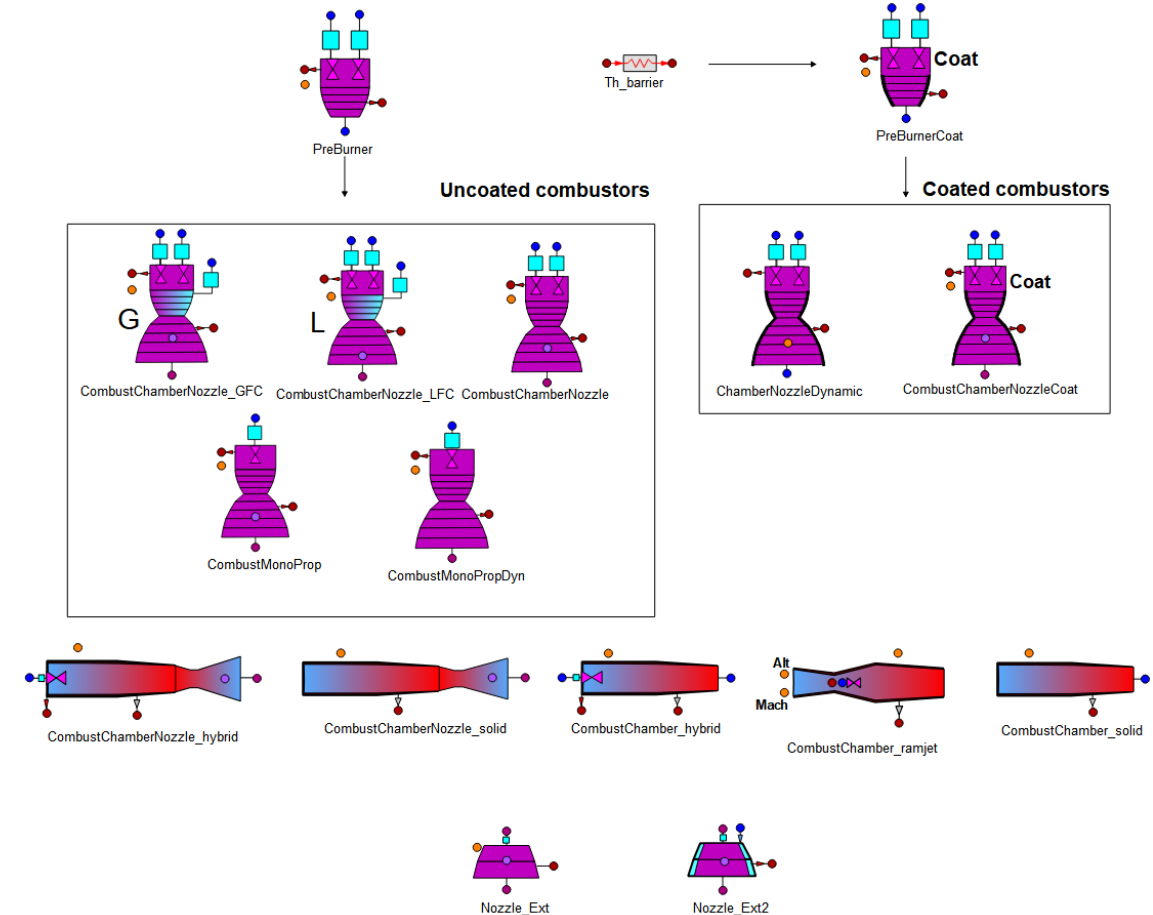
○ Discretized Channel with Wall

- Internal Wall, External wall and rib
- Simplified
 - One thermal node per part of the Wall)
- Normal
 - 3D discretization of the wall (ns,nr and Channel Nodes)
- Quasi 2D
 - 2D discretization of the channel, with 3D wall
- Toroidal
 - 3D discretization of the wall with Cavity with 3 fluid ports at inlet and outlet



• Comb Chambers

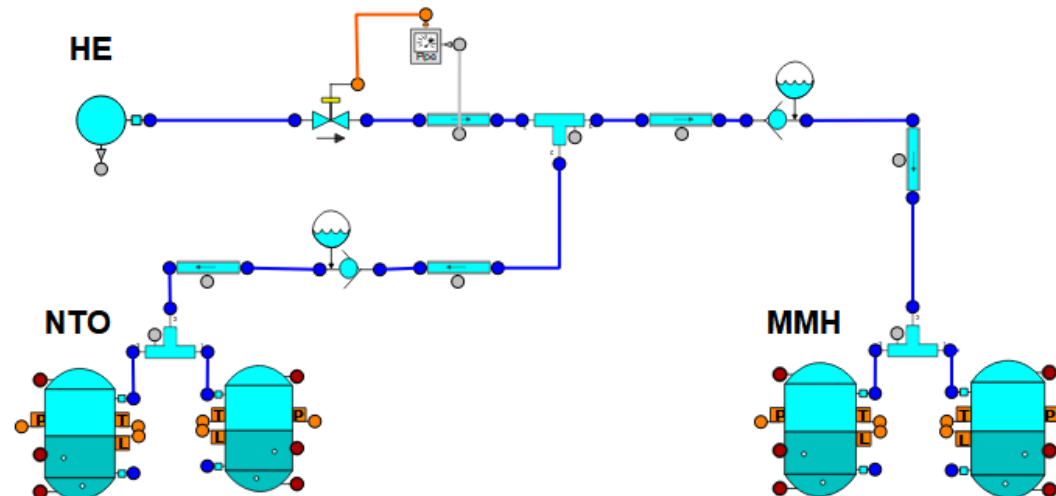
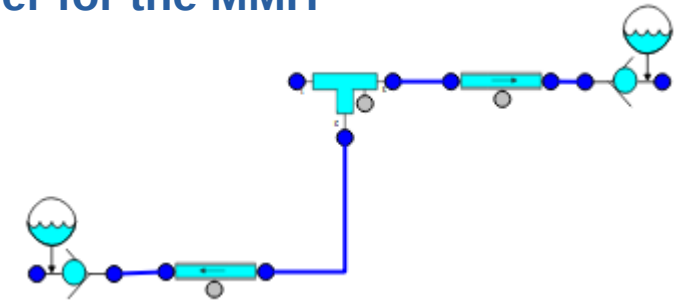
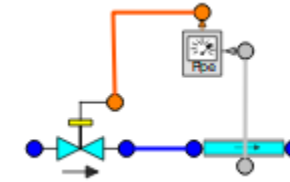
- Multiple Combustors Available
- Preburners
- Chamber + Nozzle
 - Solid/Hybrid
 - Ramjet
 - MonoPropellant
 - Dynamic Mesh
 - Coated/Uncoated
 - Liquid/Gas film cooling



- Different options for heat exchange, vaporization and combustion

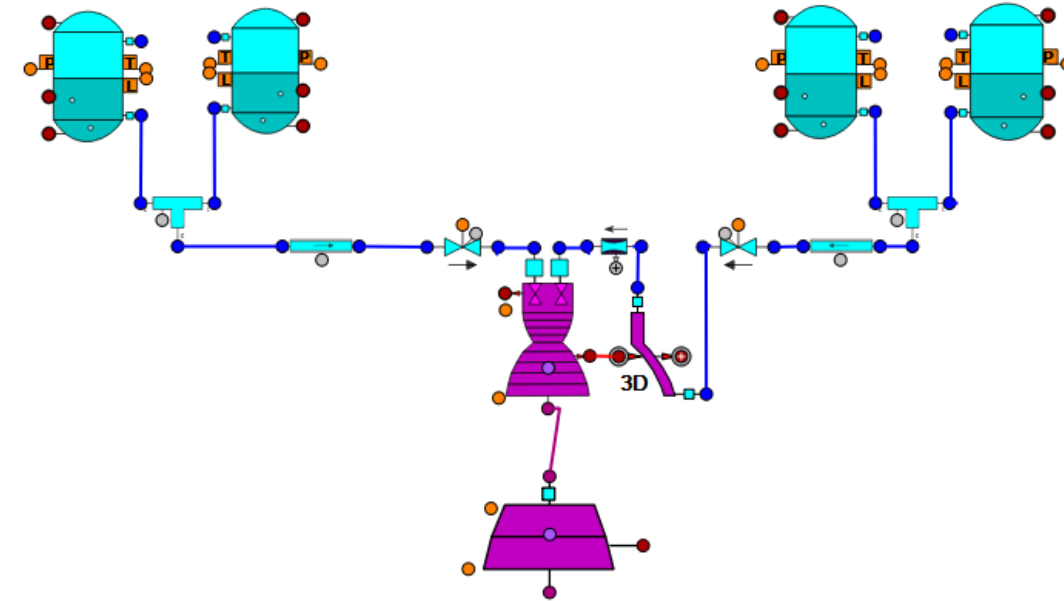
- Tank Pressure Fed

- Constant Pressure in the MMH/NTO TANK during operation
 - Option A. Pressure Regulator Down
 - Option B. Valve with user defined control (PI, PID...)
- Pressure Regulator acting as discharge valve of the He Plenum
- The flow is separated in two different lines, one for the NTO and another for the MMH
 - Use TEE
 - Lines to simulate the pressure drop due to friction
 - Tank Check Valves to allow having two different downstream fluids, with the same pressurizer
- The flow is separated again for each individual tank, with tee



- **Combustion**

- Tee for collection of flows
- Pipes for piping
 - Adiabatic
- 3D CoolingJacket
 - Best results in accuracy/runtime
- Valves
 - Direct Control of the injection times
- CombChamberNozzle
 - Simplest component for the combustion
 - Includes Injectors and Plenums
- NozzleExt
 - Nozzle without CoolingJacket



• Thermal Elements

○ Capacitive elements

- Calculate evolution of temperature

$$C \cdot \frac{dT}{dt} = \sum q_i$$

- 0D components

- DNode, BNode

- 1D components

- Cylinders, DNodeVec, Wall, Saddle, Plate

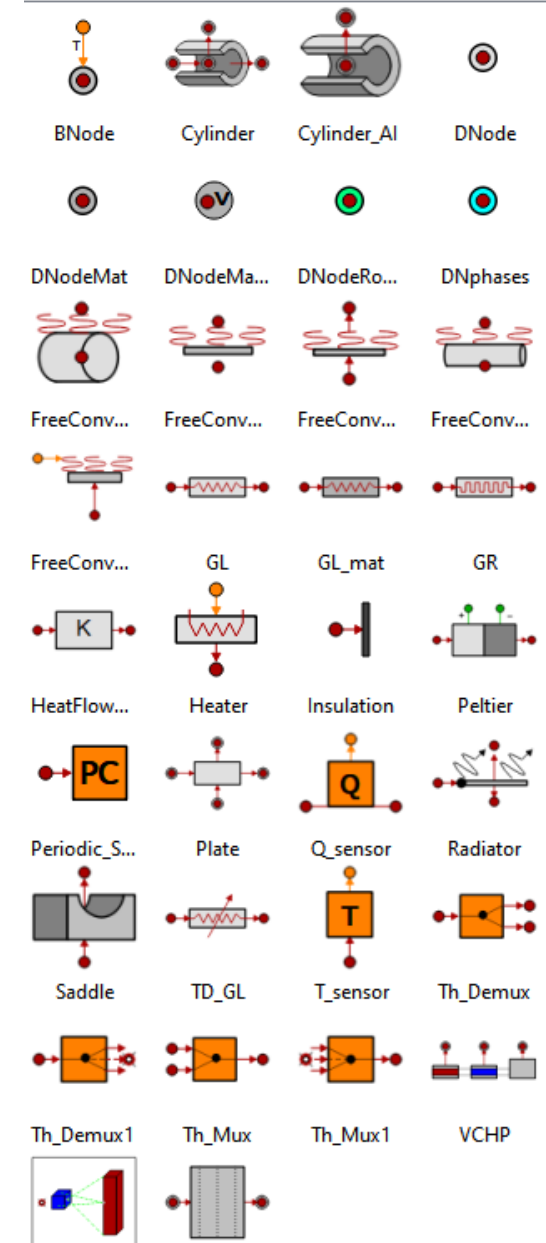
○ Resistive elements

- Calculate heat rate

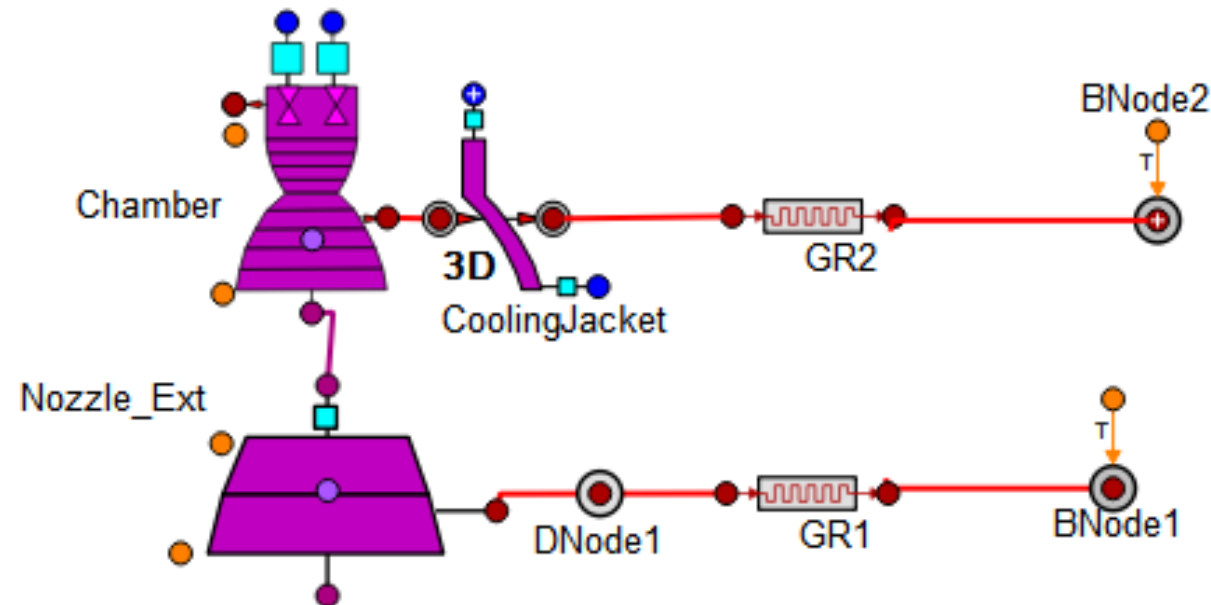
$$q = f(T)$$

- Conduction, Radiation, Natural Convection

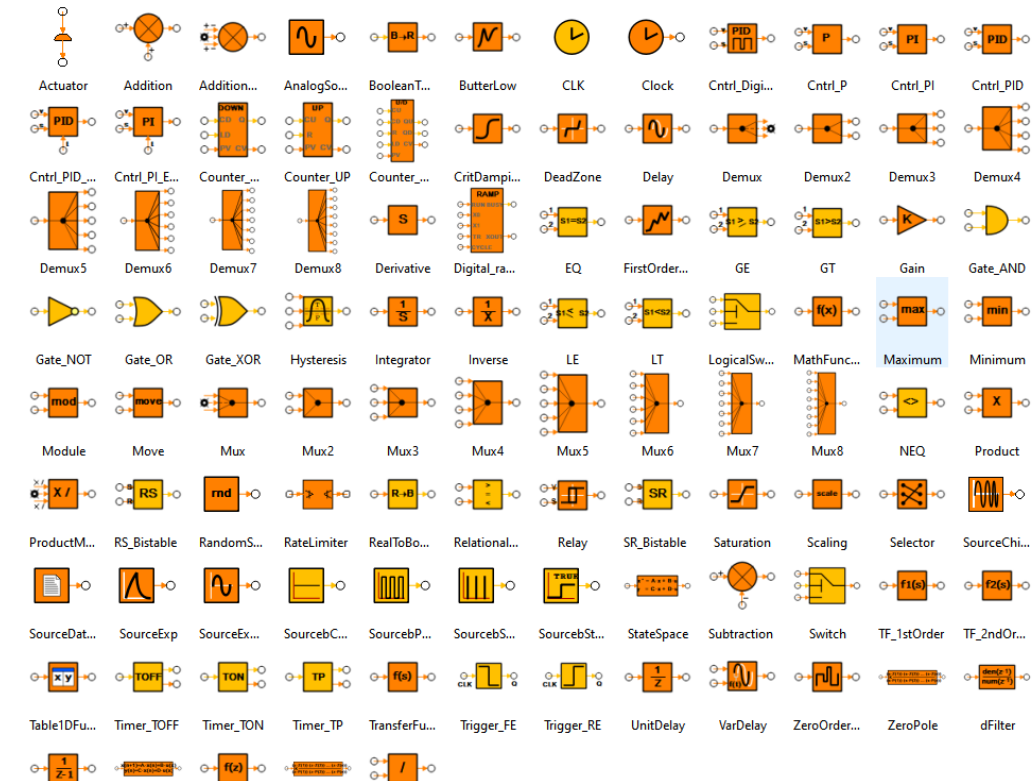
- GL, GR, FreeConv



- Radiation between the nozzle walls and the ambient
 - Nozzle Wall
 - Considered inside the Cooling Jacket component
 - For the exterior Nozzle, a DNode is used
 - Radiation considered
 - GR component
 - Ambient Modelled as a constant Temperature
 - Use BNode

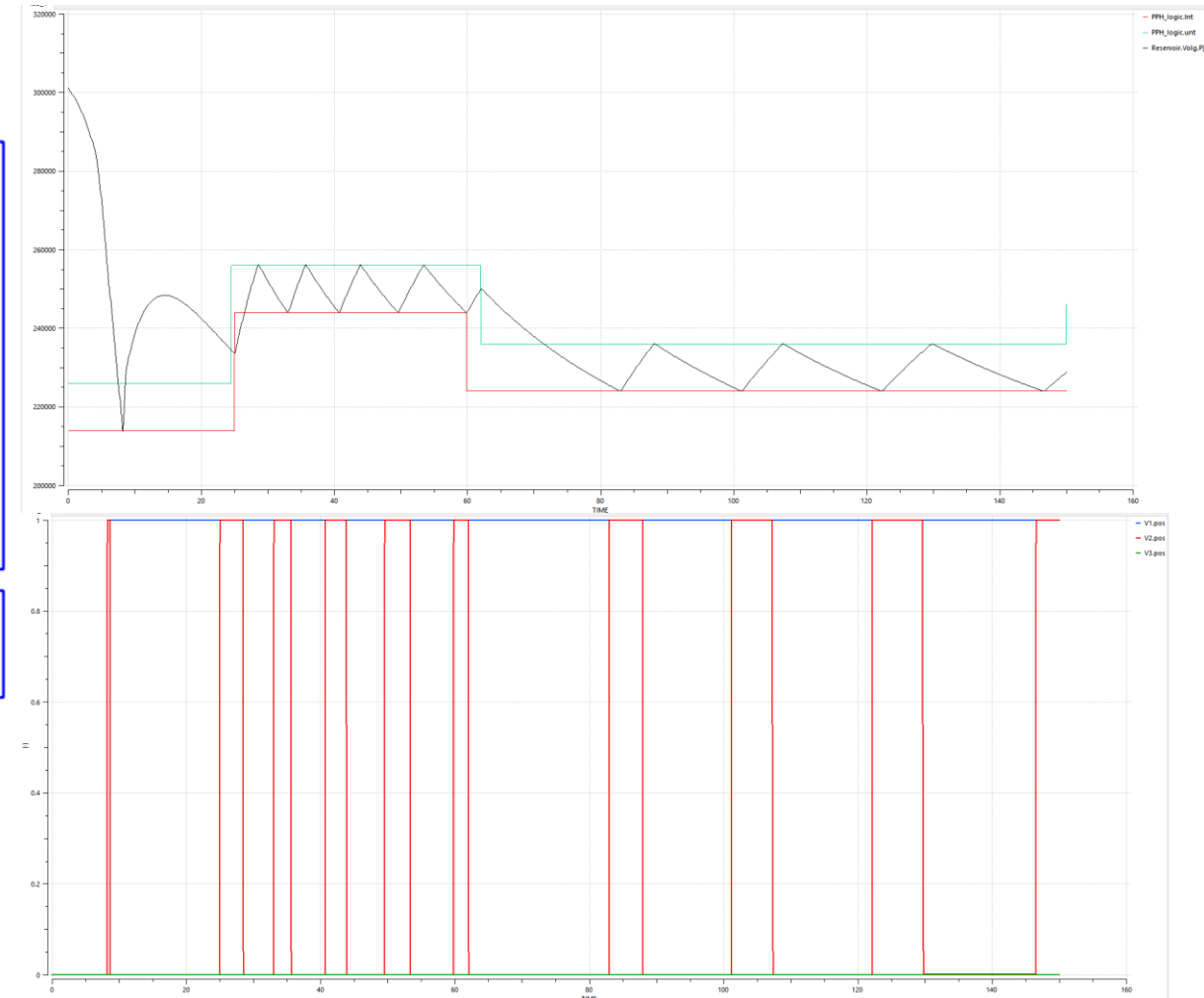
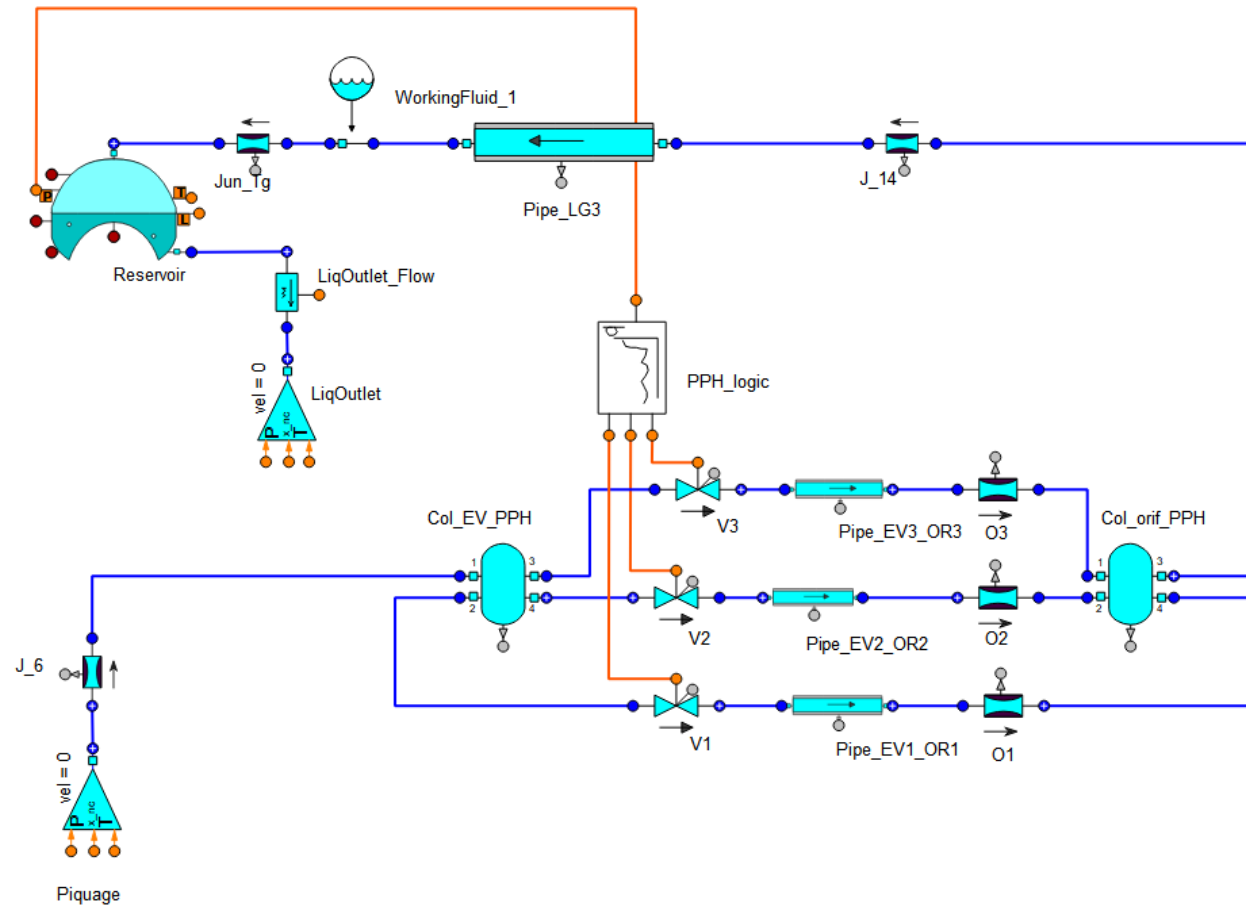


- Control Elements
 - Extensive library with analog and digital control elements
 - Most common components used in ESPSS
 - Analog Source
 - PI, PID Controller
 - Analog operators
 - Transfer Functions
 - For the example no specific control will be used
 - Valve Positions will open and close at a given TIME
 - Defined in Experiment



Control Elements

- If required, the user can create components for tailored control
 - Combination of control components and use of discrete statements



Setting Up Simulations. Initialization

FLUID_FLOW_1D

Building a model (III)

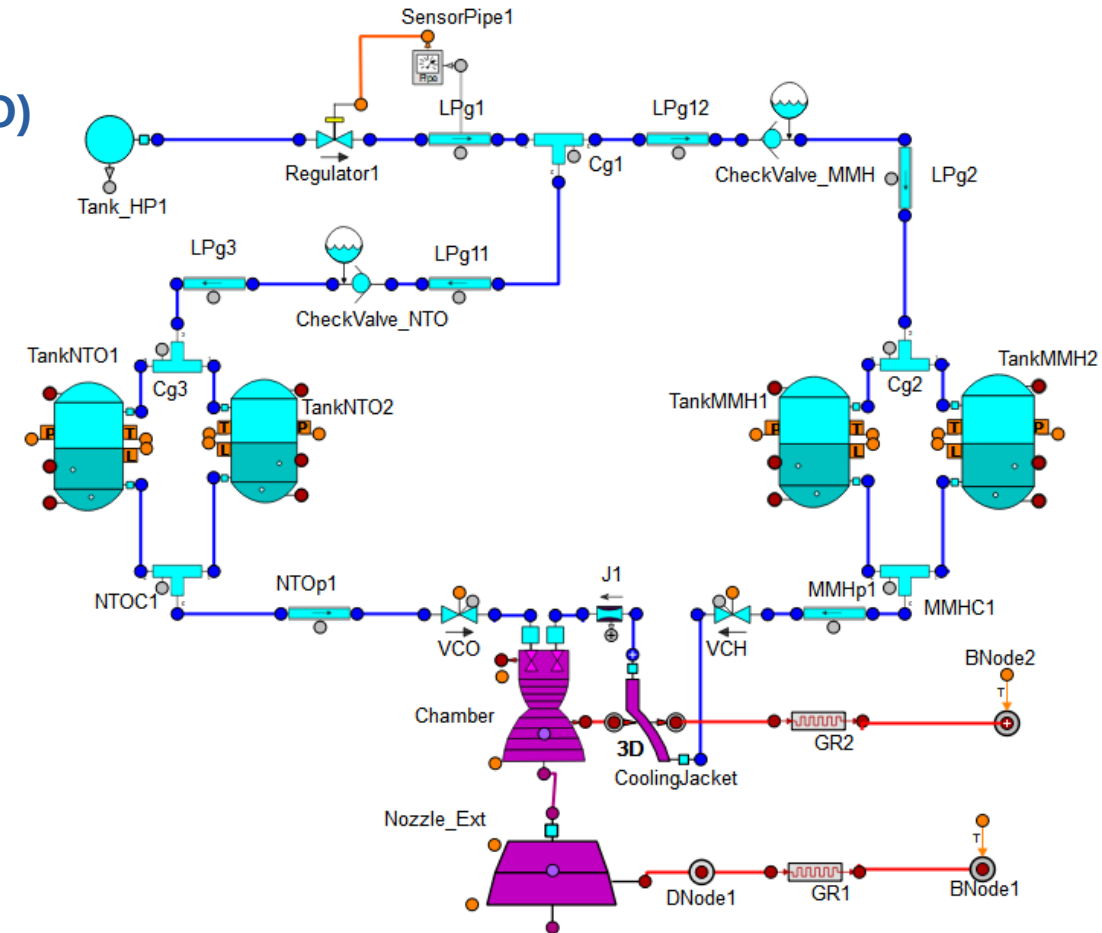
- Global data variables (pressure and temperature) are useful for initializing different components to the same conditions.
- System coordinates relative to a body axis are only given in the junction-type components.
- Choosing the appropriate boundaries and initial conditions is an important point in view of the many possible fluid dynamics solutions.

Option	Case	1st Variable	2nd Variable	3th Variable
INIT_PT	Pure fluid	pressure	temperature	--
	Two fluids	pressure	temperature	Non-condensable mass fraction
INIT_RT	Pure fluid	density	temperature	--
	Two fluids	density	temperature	Non-condensable mass fraction
INIT_Px	Pure fluid only	pressure	quality	--
INIT_Tx	Pure fluid only	temperature	quality	--
INIT_RPT	Two fluids only	density	pressure	temperature

INITIALIZING FLUID CONDITIONS

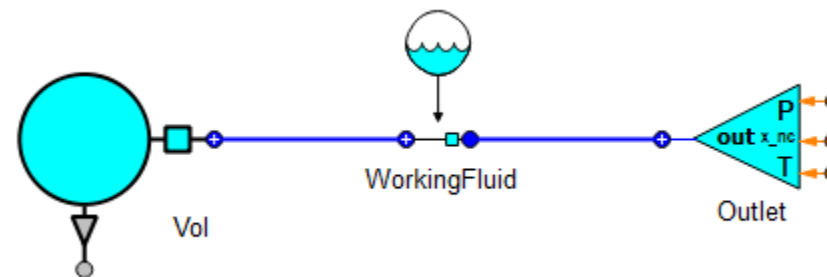
- It is essential to provide a good initialization of the fluid conditions of a model in order to avoid difficult and unphysical transients at the beginning of the simulation: care has to be taken not to impose mass flows not compatible with the sonic limitations or inappropriate P/T conditions
- A good practice is to initialize a whole branch (e.g., line between closed valves) at the same conditions
- The use of model global data is useful to initialize different components at the same conditions (e.g., pressure and temperature)
- It is advisable to initialize with zero mass flow when possible ($m_o = 0$) since a non-zero initial mass flow can force high initial accelerations
- Capacitive components can be initialized with a fraction of non-condensable fluid, which allows studying easily gas mixtures or gas sweeping processes: initialize non-condensable mass fraction to 1 for those components containing gas, 0 for those containing liquid (or main fluid) and a value in the middle for mixtures

- Define Global Initialization
 - HP (Helium Tank)
 - 200e5 Pa
 - PgO,TgO (Everything between Regulator and VCH/VCO)
 - 15e5 Pa, 275 K
 - Pext,Text (Below VCH and VCO)
 - 1000 Pa, 270 K
 - x_nc
 - 0 Upstream Check Valves
 - 1 downstream check valves for Helium
 - 0 Between Tanks and Injection Valves
 - 1 for components with P = Pext
 - z_jun
 - 0 downstream tank, z_reg for Helium
- Type of Initialization
 - INIT PT
 - All Components



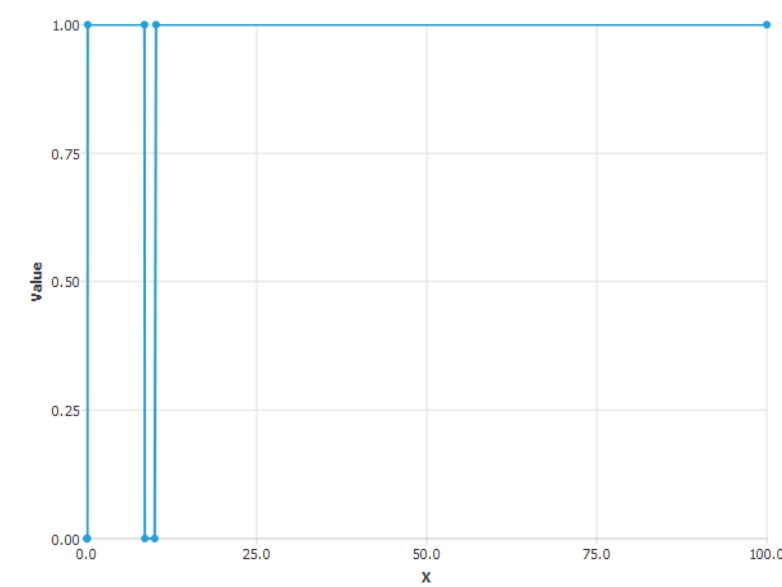
BOUNDARY CONDITIONS

- Boundary components will determine **pressure / temperature / density / nc mass fraction** at the model interfaces
- The boundary values can be time dependent
- Boundary values can be defined connecting CONTROL components to the boundary ports; if those ports are left unconnected the boundary conditions will be automatically included in the experiment
- Care must be taken to choose the appropriate boundary component



Boundary Conditions Imposed

- VCO open at 0.1. Close 8.5. Open 10 to end
 - $\tau = 0.05$
- VCH open at 0.15. Close 8.2. Open 10 to end
 - $\tau = 0.05$
- Ignition can start at 0.2. Off 8.3. On 10.2 to end
 - $\tau = 0.001$
- Starter ($m = 0$, not considered)
- Nozzle $P = P_{\text{ext}}$
- Ambient Temperature = T_{ext}
- Tanks Adiabatics
- No Condensation in tanks (dropletDrop_speed = 0)





PIPES – TUBES

- Use the Pipe or Tube components to simulate a 1D line discretized in several nodes. Pipe components include the modeling of the wall assuming one thermal node per fluid node
- As a general rule, we advise to use few (1 to 5) nodes per pipe if the user wants only to simulate the global pipe inertia and pressure drop effects.
- The user can increase the number of nodes for obtaining particular results at a given distance inside the Pipe. Please note that the use of very small pipes can give rise to numerical problems (time step is proportional to the smallest length of Pipe)
- The calculation of the heat exchange coefficient in pipes takes into account two-phase regimes. There may be some problem when passing near the critical. Then some numerical correlations using Cp risk of having discontinuities. If it is the case (the simulation becomes slow), it is advised to use `ht_option = HT_critCond` while this situation lasts



COMBUSTION CHAMBERS

- Use a SET_OF to only consider the relevant chemicals

```
COMPONENT PressureFedEngine_FR ( SET_OF(FLUID_PROPERTIES.Chemicals_full) BG = { H2O, N2O4, MMH, N2, CO2, He } )
```

- Create Construction Parameters in schematic for the number of subsonic and supersonic nodes
- Ideal Nozzle formulation is more robust
- Select the proper component for the simulation based on the physical effect
- If using Finite Rate, try to set the first node of the combustion with equilibrium to stabilize the combustion
- Properly set MR_min, MR_max, MR_min_arrhenius and Ignit_Flag boundary, to avoid calculating erroneous combustions



GLOBAL VARIABLES

- Global simulation parameters automatically appear in the **experiments** to control the integration; the most relevant are:
 - **IMETHOD** (integration method): it is advisable to use IDAS_SPARSE or CVODE_BDF_SPARSE to save CPU time; DASSL_SPARSE and the corresponding dense version of all them can also be used
 - **Gravity** (GRAV variable relative to the body axis used to define the Junction elevations) is the “vertical” body acceleration
 - **Damp** represents the artificial viscosity parameter; values from 0.1 to 0.5 provide good smoothing of numerical pressure oscillations, values in the range 0.5 – 1.0 slightly smooth water-hammer pressure peaks, but saves CPU time. Use Damp = 1 in priming cases to account for great turbulence
 - **REL_ERROR** and **ABS_ERROR**: the absolute and relative error permitted in the integration
 - Their value depends on the integration method and special conditions, but should be in the range 1e-4 to 1e-7
 - New models may use values around 1e-5 and then try smaller or bigger values depending on the results and stability of the model
 - Usually, the calculation is stable and fast by setting these variables to 1e-5

NUMERICAL LIMITATIONS

- All the **fluid ports** of a model should be connected; otherwise, the default partition (mathematical arrangement of equations of the complete model made automatically by EcosimPro) could contain inappropriate variables as boundary conditions
- Other ports such as control, signals or even thermal are allowed to remain disconnected (even though the latter are recommended to be connected): the open ports containing inputs will appear as boundaries that should be defined in the experiment
- Direct steady conditions are difficult to obtain using the STEADY command on a transient model: it is better to use the STEADY library or let the solver obtain stationary conditions
- The **Jun_TMD** component (imposed mass flow) should only be used as an exit or inlet boundary condition and not in the middle of a network, since it will act as a pump but assuming continuity in enthalpy and this could lead to numerical issues
- Time-dependent boundary conditions should not force instantaneous mass flow changes because this implies infinite fluid acceleration
- A characteristic actuator time of 0 in **Valves** (input data “tao”=0) may produce unphysical pressure peaks



NUMERICAL LIMITATIONS

- Library components have been designed to deal with homogenous two-phase, two-fluid conditions: in the situation of a high pressure liquid front traveling along a pipe filled with gas at low pressure (priming cases), the pipe discretization can produce some numerical pressure peaks; nevertheless, the main value of the pressure surge and its reflections are well calculated with relatively few nodes
- The simulation of a mixture process (priming) between the liquid and the gas slightly depends on the number of nodes, especially in the situation of pipes filled with a mixture of liquid and non-condensable gas: the flow in these pipes may be no longer 1D, but like in a sloshing volume, with low speed and high pressure fluctuation due to the abrupt stops of the sloshing flows against the closed ends
 - Setting the “ufm” option (an input for each Pipe) to damp_mod, the Damp global parameter is automatically multiplied by $k_d = L / \Delta x / 10$: this way, the influence of the number of nodes is minimized



NUMERICAL LIMITATIONS

- Sudden expansion of a liquid under the saturation pressure could produce very low temperatures, even under the triple point (snow formation): this dangerous situation is well supported by ESPSS by assuming the vapor phase is extrapolated as a real gas at $P < P_{\text{triple}}$, $T = T_{\text{triple}}$ and that the liquid behaves as a saturated liquid at T_{triple}
- Under priming in vacuum conditions it is preferable to initialize Pipes/Tees with condensable vapors at a pressure lower than the saturation one, rather than with non-condensable gases; use non-condensable gases if it is really necessary
- If the simulation runs slow, try the following:
 - Check initial conditions (pressures 100 → 1000 Pa for example)
 - Check if there is any discontinuity
 - Check number of nodes
 - Check the integration method
 - Increase ABS_ERROR and REL_ERROR integration parameters (but never above $1e-4$)



SOLVER	RUNTIME (S)
CVODE_BDF_SPARSE	113
IDAS_SPARSE	192
DASSL_SPARSE	251
CVODE_BDF	576
IDAS	900
DASSL	1500

REL/ABS_ERROR	RUNTIME (S)
1e-4	NOT_OK
1e-5	NOT_OK
5e-5	NOT_OK
1e-6	165
1e-7	250
1e-8	400

DAMP	RUNTIME (S)	Pressure Peak (Bar)
0.1	800	245
0.3	269	184
0.5	179	183
0.8	141	168
1	85	166
1.3	65	153
1.5	61	148

P_Ext	RUNTIME (S)
10	NOT_OK
100	120
1000	115
10000	106

Debugging

Running a Model for the First Time

- Below some Monitor inspections the user should make when running a model for the first time or when having trouble:
 - Use REPORT_MODE = IS_STEP in the experiment
 - Verify the coherence of the inputs in the Monitor tool: e.g., valve/junction areas (*.Ao), tube lengths and diameters (*.L, *.D), fluid volumes (*.Vo), tank levels (*.level), etc.
 - Verify the initial conditions: plot Pressure, Temperature (*.P, *.T) and non-condensable mass and void fraction (*.x_nc, *.alpha) values at the relevant fluid volumes and tees
 - Do the same for relevant pipes, 1D tanks and combustors (*.T[*], *.P[*], *.alpha[*], etc.)
 - Plot the mass flow values (*.m) at relevant junctions/valves and the heat fluxes (*.q[*]), and verify the values
 - Plot valve positions (*.pos) and boundary conditions (*.P, *.T, *.x_nc) at Volxx_TMD type components, and verify that they behave as defined in the BOUND block of the experiment

Detect Simulation Issues

- Check that the initial conditions of all the components are coherent and consistent
 - Set the same initial pressures, temperatures and non-condensable mass fractions for all the components connected in a branch without restrictions (e.g. between valves)
 - Check that the elevations of the junctions are good (“z_jun” variable)
- Check if there is any strong discontinuity that is making the simulation slow down at a certain time, and try to smooth it
- Check the integration method (IMETHOD variable on the experiment file) and try using IMETHOD = IDAS_SPARSE or IMETHOD = CVODE_BDF_SPARSE
- Check if the fluid phase is changing continuously at some point in the circuit (examine “alpha” variable) and analyze if it comes from a bad initialization, then revise initialization
- Avoid excessive discretization (too many nodes)
 - A good value could be $1/5 \cdot \text{length/diameter}$
 - It can be increased in zones where a specially detailed data profile is needed (Priming, WH)

Detect Simulation Issues

- If any problem remains, plot the derivative variables (*) and analyze numerical oscillations. If this is the case, it means that you probably have some hidden discontinuities, perhaps in the boundary conditions, or calculating properties (fluid conditions out of the scope?), or when detecting a ZONE change, etc. Try to change some input data and options such as:
 - Changing the number of nodes of the pipes
 - Increasing Damp parameter and reducing ABS_ERROR, REL_ERROR
 - Trying equivalent simplified topologies or boundary conditions, pipe diameter & lengths, heat exchange correlation options, etc.
- Other tips that may help if the model crashes at some point:
 - Set command ***setStopWhenBadOperation(TRUE)***
 - Write it in the BODY of the experiment (before calling the integration)
 - This option detects bad operations (division by zero, invalid power, etc.) and stops the simulation telling which variable is invalid

Detect Simulation Issues

- Other tips that may help if the model crashes at some point:
 - Use ***produceLogAsciiFile()***
 - Write it in the BODY of the experiment (before calling the integration)
 - This option produces an extra output file named ***simulation.log.txt*** with information on the value of each variable in the same order as it is calculated and at each call to the residues function (FRES()), the one calculating all the model variables and residues).
 - It has the following optional parameters

```
NO_TYPE produceLogAsciiFile(    INTEGER fromCall=-1,    INTEGER toCall=-1,  
                                REAL fromTime=-1.0,    REAL toTime=-1.0 )
```
 - Use ***eResiduesLogEnable()***
 - Write it in the BODY of the experiment (before calling the integration)
 - Generates lot of information in transient experiments, consider enabling it during certain timesteps of the simulation.

VARIABLE NAMES

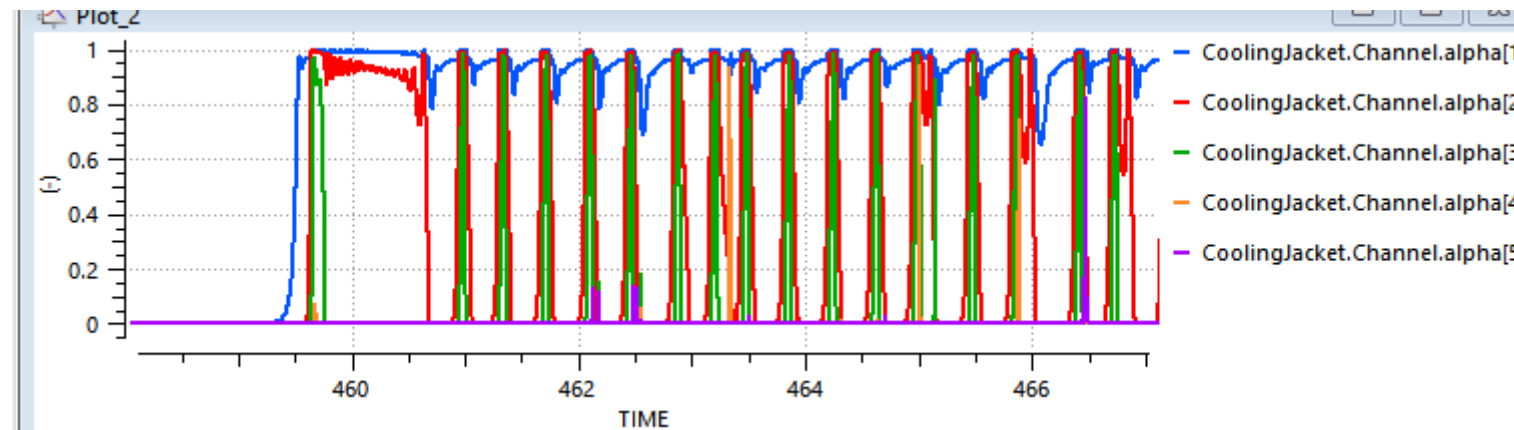
- How to find a variable to exploit results

VARIABLES	Monitor Filter	Component Type	VARIABLES	Monitor Filter	Component Type
Pressure (stat.) (Pa)	*.P	Volume, Tee, Cavities	Heat Coeff. (W/m2/K)	*.hc	Volume, Tee, Cavities
	.P[] dP_loss	Pipe, Tanks & Comb.		*.hc[*]	Pipe, 1D Tanks & Comb. Chambers
Temper. (stat.) (K)	*.T	Volume ...	Density (kg/m3)	*.rho	Volume ...
	.T[]	Pipe, Tanks & Comb.		*.rho[*]	Pipe, 1D Tanks & Comb. Chambers
Total & NCG mass flows (kg/s)	*.m, *.m_nc	Valve/Junction	Heat flux (W)	*.q	Volume, Tee, Cavities
	.m_jun[]	Pipe, Tanks & Comb.		*.q[*]	Pipe, 1D Tanks & Comb. Chambers
Pressure drop (Pa)	*.dP_loss	Junctions & Pipes (see note)	Friction coeff.	*.fr[]	Pipes
(ρv) _{lam} (ρv)	*.G _{lam} *.G	Valves, Junctions	(ρv) _{crit}	*.G _{cr}	Valves, Junctions
Stem pos. (0-1)	*.pos	Valves			
Non-condens. mass fraction	*.x_nc	Volumes, ...	Vapour mass fraction	*.x	Volumes, ...
	.x_nc[*]	Pipe & Tanks		*.x[*]	Pipe & Tanks
Diluted gas mass fraction	*.xd_nc	Volumes, ...	Void Fraction	*.alpha	Volumes, ...
	.xd_nc[*]	Pipe & Tanks		*.alpha[*]	Pipe & Tanks
Velocity (m/s)	*.vel	Volume, ...	Reynolds no.		Pipe, 1D Tanks & Comb. Chambers
	.vel[]	Pipes, 1Dtanks		*.Re[*]	

VARIABLES	Monitor Filter	Component Type	VARIABLES	Monitor Filter	Component Type
Mach number	*.Mach	Volumes, ...	Mach number	*.Mach[*]	
Wall temper.	*.t_wall		Wall temper.	*.t_wall[*]	
Level	*.level	Tank single 1D tank	Evap. mass flow	m_evap	1D tank
	*.Lh		Boil mass flow	m_boi	
Power	*.power	Turbo-machinery	Abs. mass blow	m_abs	
Torque	*sh*.T		rad/s	*.omega	Turbo-machinery
Wall temper.	*.t_wall*		rpm	*sh*.n	
Chemical mass fractions	bu_eq[*].x[*]	Combustors & Pipes	Reactant mixture's mass fractions	bu_in[*].x[*] or x_mix[*,*]	Combustors. (Current comp. in rate model)
	x_chem[*]	Volumes			
	noz_conc[*].x[*]	Nozzles	Temp., Pres. Mach	*_noz[*]	Nozzle
Liquid mass fractions	*.x_lf[*], *.x_lo[*]	Combustors	Thrust	*.thrust	Nozzles
Pipe end vars	*1, *n *.Tup	See notes			

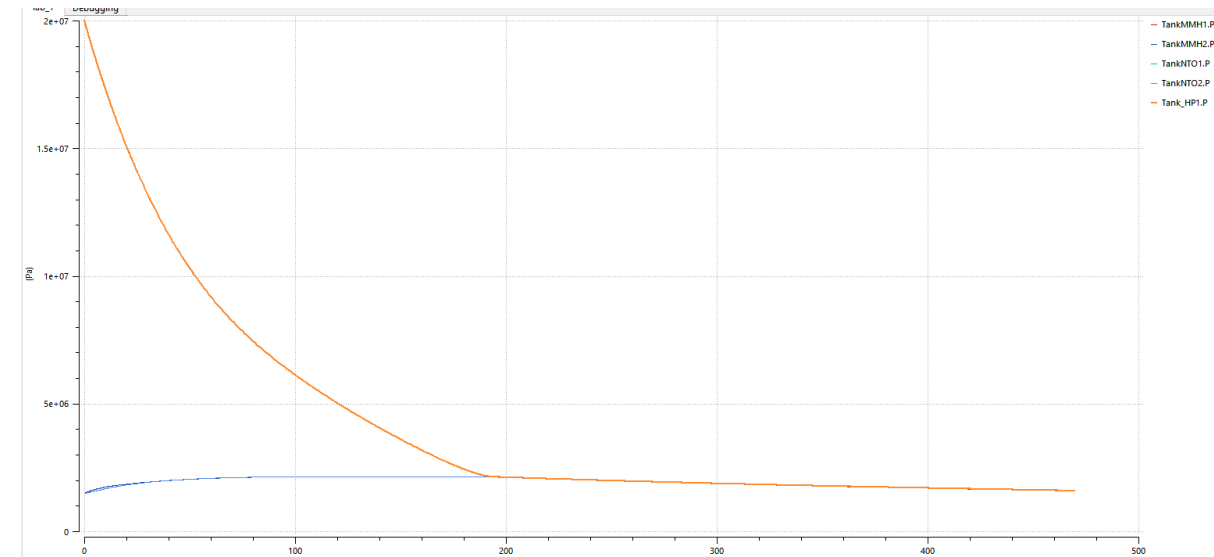
EXAMPLE

- PressureFedEngine_FR
 - We want to run the simulation until we discharge the tanks.
 - Change CINT and TSTOP
 - Simulation gets stuck at $T = 460$ s
 - Test, Change Tol, DAMP
 - No Effect
 - See the origin of the problem
 - Plot all the dynamics (*)
 - Cooling Jacket changing phase all the time



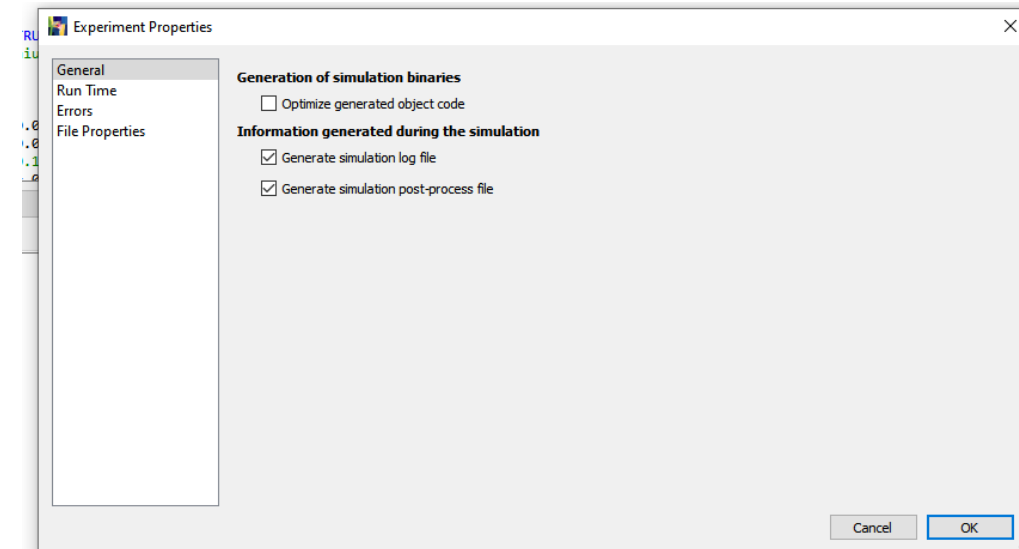
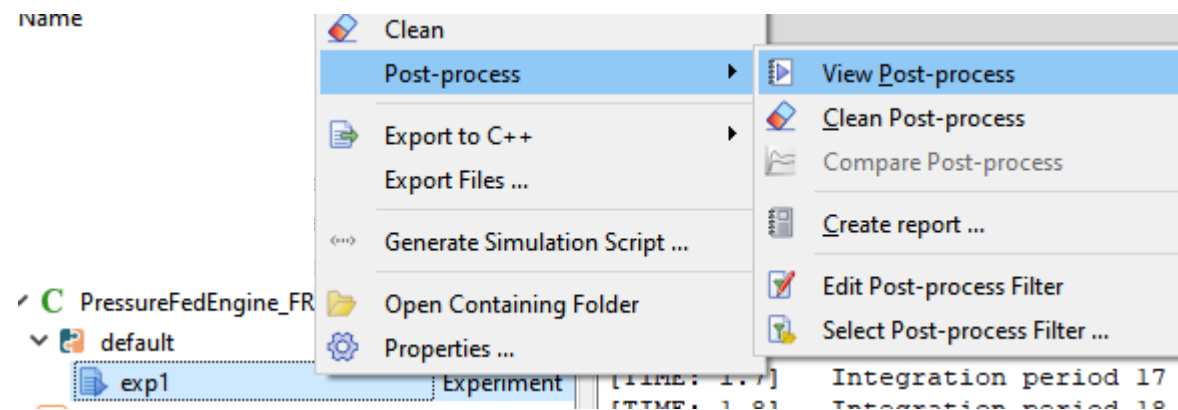
EXAMPLE

- Cooling Jacket Change phase
 - Physical Effect?
 - Interaction Cooling Jacket with Main Chamber
 - Pressure in Tanks goes down
 - Intended?
 - Bad Simulation or Control Set-Up?
 - Valves designed at Controlled Pressure
 - If Pressure Decreases, MR and mass-flow no longer the design value,
 - Solutions
 - Changing the Volume of the Pressurizer, to ensure $P_{EOL} = P_{controlled}$
 - Changing control of valves to ensure constant MR and MassFlows
 - Changing Heat Transfer Option of Cooling Jacket or Chamber
 - In this case *hc_correll* Bartz changed to *Pavli*
 - Heat Transfer is lower, the oscillations do not appear.



Post Process

- For long simulations, possibility to create binary file.
 - Properties of Experiment. Generate PostProcessFile.
 - When running experiment, file will be generated
- Possibility to access the binary instead of running simulations

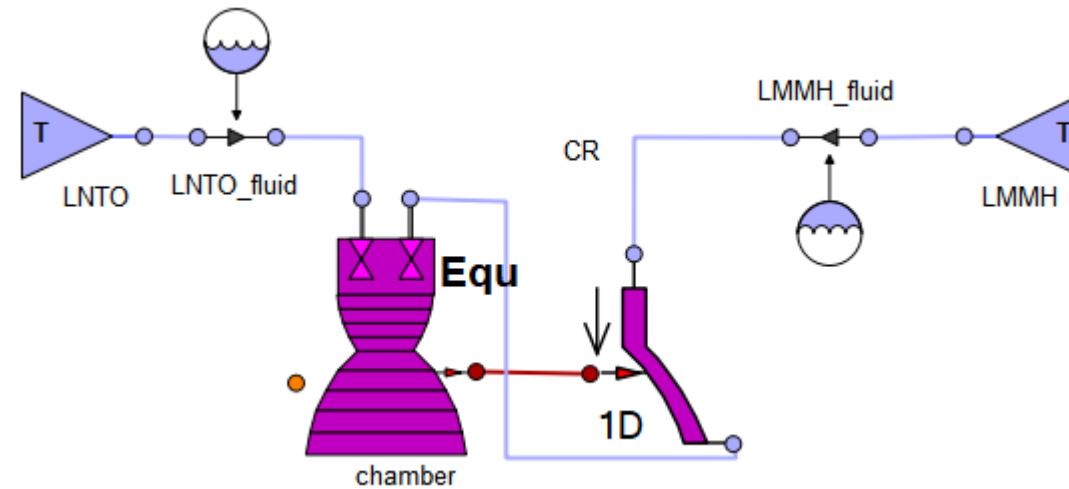


- Same Results loaded, without having to run the simulation. Can be key for long time simulations.
- Binary file can get really heavy, filter of variables can be created.

Other Analysis

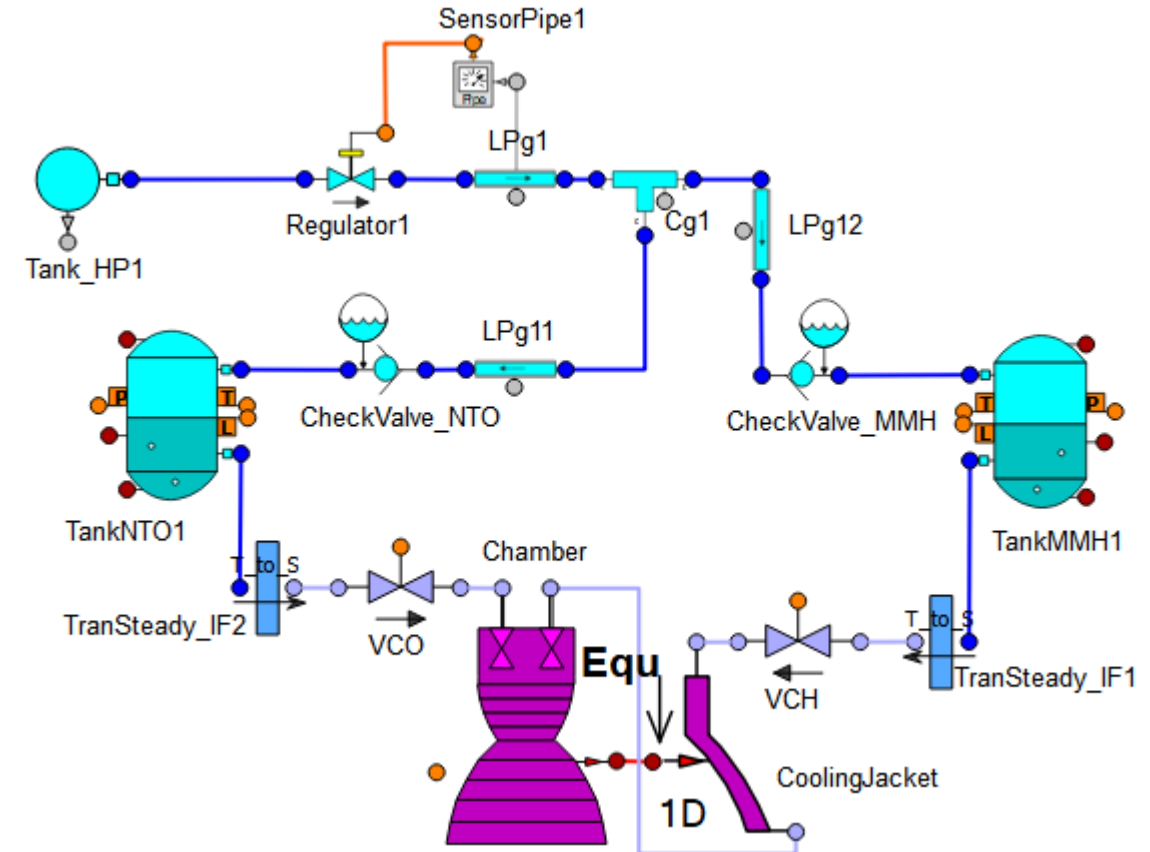
Changing the components may change the scope of the simulation

- Steady Formulation for the Combustion Chamber
- Constant Feeding Pressure
- Design CJ and Chamber



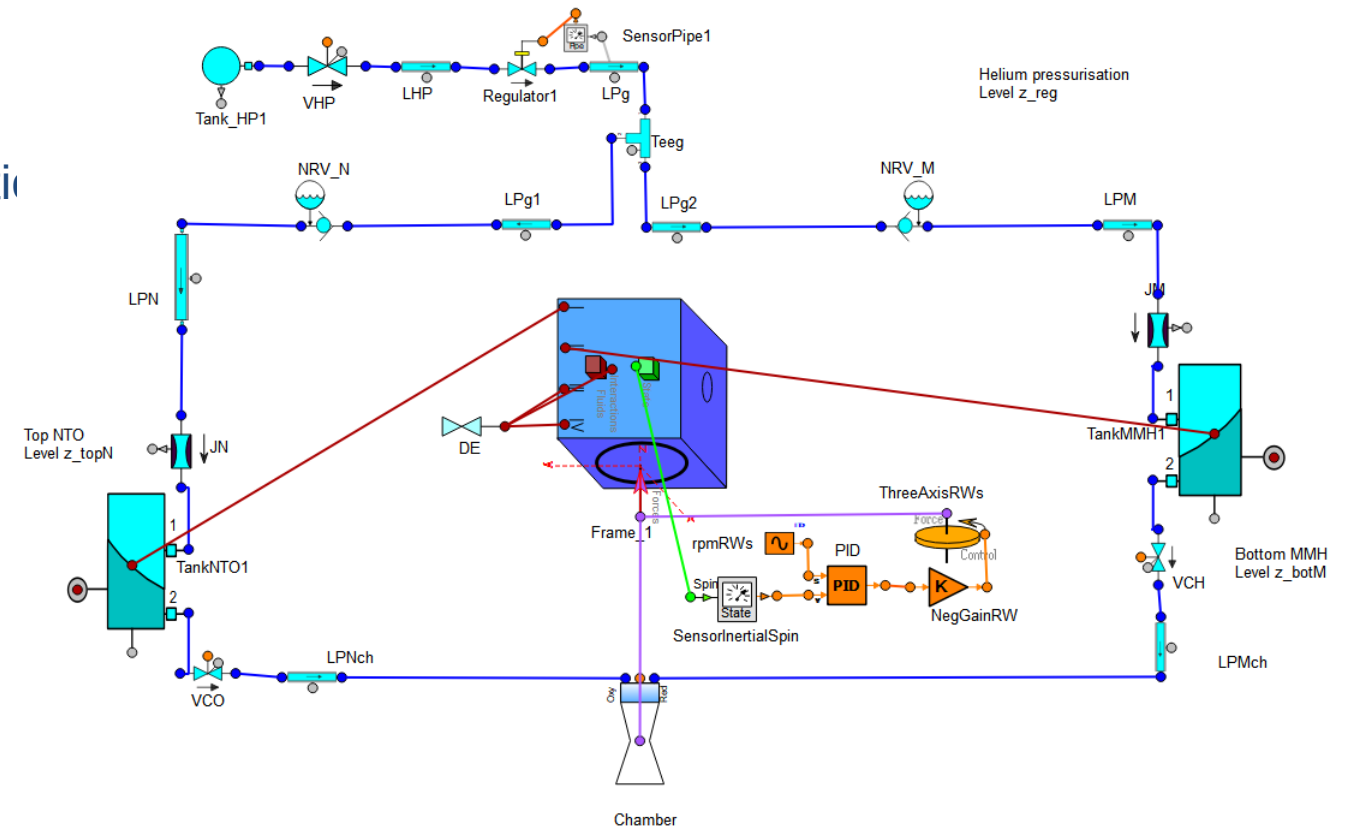
Changing the components may change the scope of the simulation

- Coupling of Pressurizer with Steady Combustion
 - Simplified Combustion
- Off Design Performance of Chamber
- Design of components at EOL



Changing the components may change the scope of the simulation

- Tanks with sloshing
 - No vaporization
- No CJ
 - Perfect Liquids
- Simplified Chamber based on empirical formulation
- Control of Satellite with RW
- Evolution of Satellite Attitude and Orbit



Conclusions

- **Keys for being proficient in ESPSS**
 - **Identifying the scope of the simulations**
 - **Select the appropriate libraries and components**
 - **Initialization**
 - **Using Global Variables**
 - **Debugging**
 - **Understanding and setting the boundaries**
 - **Key Variables**
 - **Using the monitor**
 - **Initialization, Physical Problems and Numerical Problems**

Contact

- Main format to contact and report bugs or improvements:

- Bugzilla

- <https://sprs.ecosimpro.com/>

- Mail

- General Information

- info@ecosimpro.com

- Support Team

- support@ecosimpro.com

- Any member of the ESPSS support team

- Maria Aranda

- maranda@empre.es

- Shankara Coello

- scoello@empre.es

- Request for improvements can influence next developments or ESA ESPSS projects.



Questions?